Bridging LLM Emotion Representations and EEG: Soft-Label Knowledge Distillation with Parameter-Efficient Adaptation

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Abstract

Recent interpretability research reveals that large language models develop internal emotion representations organized along the same valence–arousal dimensions found in human psychology. We introduce **EMOTIONKD**, which exploits this brain–LLM alignment: we extract emotion vectors from instruction-tuned LLMs and use their relational geometry as soft-label knowledge distillation targets for EEG emotion classification, teaching the model that *anger is closer to disgust than to joy* rather than treating all misclassifications equally. On FACED (9-class, 123 subjects), EMOTIONKD achieves $\mathbf{0.628}$ balanced accuracy, improving over the previous state of the art (REVE, NeurIPS 2025) by $+\mathbf{11.2}\%$ relative, with cross-subject generalization confirmed at $p < 0.001$. We validate the framework along three orthogonal axes: (i) *cross-backbone*, where KD significantly improves both CBraMod ($+\mathbf{5.2}\%$, $p < 10^{-4}$) and LaBraM ($+\mathbf{2.6}\%$, $p = 0.025$); (ii) *cross-dataset*, where with sufficient temporal granularity KD also significantly improves SEED-V on LaBraM ($+\mathbf{1.85}\%$, $p = 0.037$); and (iii) *cross-LLM-family*, where 11 instruction-tuned models from Qwen2.5 and Gemma-4 produce nearly identical emotion geometries (within-family RSA $\rho \approx 0.89$, cross-family $\rho = 0.82$), with linear cluster purity reaching 100% for positive vs. negative valence. Through Centered Kernel Alignment (CKA), we show that KD increases the similarity between EEG features and *every* LLM in our suite, including LLMs from a different family from the training teacher, indicating that KD absorbs a universal emotion structure rather than overfitting to one specific LLM. A key architectural finding is that freezing the pretrained backbone and adding lightweight adapters (310K parameters) matches full fine-tuning (4.9M parameters): the critical factor is the *quality of supervision*, not model capacity. We release our code, emotion vectors, and pretrained adapters to advance brain–language alignment research for EEG.

1 Introduction

How do the brain and large language models represent emotions—and can their shared structure be exploited? Emotions in human psychology organize along continuous dimensions of valence and arousal [1], a geometry that neuroscience has shown is reflected in distributed neural representations. Recent interpretability research reveals a striking parallel: LLMs develop internal emotion representations with the same dimensional structure, emerging naturally from language modeling without explicit emotional supervision [2, 3, 4]. This convergence suggests a deep alignment between how biological and artificial neural networks encode affect—an alignment we propose to exploit for EEG-based emotion recognition. While EEG foundation models such as CBraMod [5], LaBraM [6], and REVE [7] have established strong baselines on emotion benchmarks like FACED [8], they treat

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emotion categories as integer indices, discarding the relational structure that both brains and LLMs naturally encode.

The standard cross-entropy loss penalizes all misclassifications equally—confusing *joy* with *amusement* costs the same as confusing *joy* with *anger*. Recent work has begun to address this: EmotionCLIP [9] reformulates EEG emotion recognition as EEG-text matching using CLIP embeddings of emotion labels, while EMOD [10] projects labels into a continuous valence–arousal space with soft-weighted contrastive learning. However, these approaches rely on hand-crafted text templates or predefined coordinate systems. We propose instead to use the *internal* emotion geometry that emerges within LLMs—a richer, data-driven representation that encodes not just valence–arousal position but the full relational structure among emotions, including within-valence distinctions (e.g., *anger* vs. *fear* vs. *disgust*).

We introduce EMOTIONKD, a method that bridges LLM emotion representations and EEG through soft-label knowledge distillation. Given any instruction-tuned LLM, we extract emotion vectors by prompting the model to generate emotion-laden stories and recording hidden state activations at intermediate layers. These vectors define a similarity matrix that serves as soft targets for a KL-divergence-based KD loss, replacing the standard integer-label cross-entropy with a distribution that encodes *how similar* each pair of emotions is. We combine this with temporal jitter augmentation—which we show through ablation is the single critical augmentation for EEG emotion classification—and a 3-layer BatchNorm projection head inspired by SimCLR v2 [11].

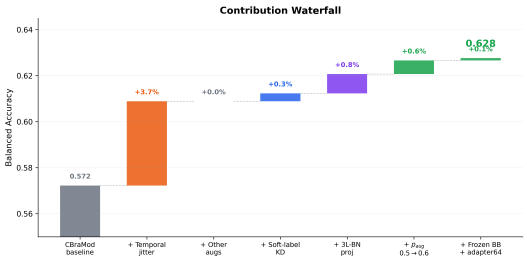


Figure 1: Contribution of each component of EMOTIONKD to the final performance on FACED 9-class emotion recognition. KD adds $+1.8\%$ over augmentation alone; adapter PEFT preserves the gain with $20\times$ fewer trainable backbone parameters.

Three orthogonal axes of validation. A core methodological contribution of this paper is to evaluate the framework along three independent axes: **(1) Cross-backbone:** we replicate KD on both CBraMod (criss-cross transformer, ICLR 2025) and LaBraM (standard ViT, ICLR 2024) and find significant improvements on both ($+5.2\%$ and $+2.6\%$). **(2) Cross-dataset:** we test on FACED (9 classes) and SEED-V (5 classes), where the KD effect on SEED-V is unlocked when input granularity is increased to 5 s windows ($+1.85\%$ on LaBraM, $p = 0.037$). This reveals a practical *token-count requirement*: KD efficacy scales monotonically with the number of EEG tokens per sample. **(3) Cross-LLM-family:** we extract emotion vectors from 11 instruction-tuned models in two families (7 Qwen2.5 sizes + 4 Gemma-4 variants), and find that within-family and cross-family RDM agreement are nearly identical (within $\rho \approx 0.89$, cross $\rho = 0.82$), confirming that LLM emotion geometry is genuinely universal across families.

A second key finding emerges from our representational alignment analysis: **Centered Kernel Alignment (CKA)** [12] between LLM features and EEG features increases for *every* LLM in our 11-model suite after KD (ΔCKA $\approx +0.03$), and the highest post-KD alignment is achieved with Gemma-4 31B—even though training used Qwen-14B as the teacher. This argues that KD does not simply pull EEG features toward the specific teacher; it absorbs a universal emotion structure that aligns with all sufficiently capable instruction-tuned LLMs.

We additionally find that **freezing the pretrained CBraMod backbone and adding lightweight bottleneck adapters (310K trainable parameters) matches full fine-tuning** on FACED, achieving balanced accuracy $\mathbf{0.628}$. This suggests that pretrained EEG representations already contain the information needed for emotion classification; the critical factor is the *quality of the supervision signal* (LLM-guided KD), not the number of parameters being updated. We validate this finding through a systematic sweep of adapter bottleneck dimensions, showing a clear scaling curve with an optimal capacity at rank 64.

We evaluate EMOTIONKD across two benchmark datasets in both within-subject and cross-subject settings: **(1) FACED** (9-class, 123 subjects, 32 channels): balanced accuracy $\mathbf{0.628}$, improving over the previous SOTA (REVE, NeurIPS 2025) by $+\mathbf{11.2}\%$ relative. **(2) SEED-V** (5-class, 16 subjects, 62 channels): with the 5 s preprocessing variant, KD significantly improves balanced accuracy and

Cohen's κ on both backbones, with bootstrap 95% confidence intervals that exclude zero. Cross-subject evaluation on FACED (5-fold leave-group-out) shows significant improvement over baseline ($p < 0.001$), confirming generalization to unseen subjects.

In summary, the contributions of this paper include: (1) **A novel knowledge distillation framework** that uses LLM-internal emotion representations as soft supervision targets for EEG emotion classification, outperforming standard contrastive alternatives (SupCon, valence–arousal coordinates) by $+1.8\%$. (2) **Cross-backbone, cross-dataset, and cross-LLM-family validation**—the most thorough validation of an EEG-LLM alignment framework to date. (3) **Empirical CKA evidence** that KD increases representational alignment between EEG and *every* instruction-tuned LLM (not just the training teacher), indicating that KD absorbs a universal emotion structure. (4) **The frozen-backbone + adapter finding**, showing that the quality of LLM-guided supervision matters more than trainable parameter count. (5) **Analysis of LLM emotion geometry universality** across 11 instruction-tuned models from 2 families (Qwen2.5 0.5B–72B and Gemma-4 E2B–31B), with detailed scale analysis and a token-count guideline for applying KD to compressed EEG representations.

2 Related Work

EEG Foundation Models. The development of large-scale pretrained models for EEG has accelerated rapidly. LaBraM [6] introduced the first large-scale EEG foundation model, pretraining on $\sim$2,500 hours of data using vector-quantized neural spectrum prediction. CBraMod [5] proposed a criss-cross transformer that factorizes spatial and temporal attention, pretrained on the Temple University Hospital EEG Corpus (TUEG). REVE [7] scaled pretraining to 60,000 hours across 25,000 subjects with a novel 4D positional encoding. CoMET [13] combined contrastive learning with masked autoencoding. EEG-DINO [14] applied hierarchical self-distillation. mdJPT [15] explored multi-dataset joint pretraining. EEG-GraphAdapter [16] proposed graph-based adapter PEFT. These models achieve state-of-the-art performance across diverse EEG tasks but treat emotion recognition as a standard classification problem with integer labels. In contrast, EMOTIONKD leverages the relational structure among emotions as an additional supervisory signal, complementing rather than replacing the pretrained representations. We additionally provide the most thorough cross-backbone validation to date by replicating our framework on both CBraMod and LaBraM and characterizing when KD does and does not generalize across architectures.

Emotion Representations in LLMs. Recent interpretability research has revealed that LLMs develop structured internal representations of emotion concepts. Sofroniew et al. [2] demonstrated that Claude's emotion vectors organize along valence–arousal axes, are causally functional (driving behavior such as reward hacking), and persist across diverse contexts. Tak et al. [3] showed through probing and activation patching that emotion processing concentrates in mid-layer multi-head self-attention, with peak encoding at 50–75% depth. Layer-wise probing of LLMs for emotion has revealed depth-dependent encoding of affective information [4], and controllable generation work has demonstrated that emotion vectors can steer text output [17]. Brain–LLM alignment studies find that scaling, but not instruction tuning beyond a threshold, increases representational correspondence with neural data [18]. We extend this line of work in two ways. First, we show that the emotion geometry is genuinely universal across instruction-tuned LLM families: 11 models from Qwen2.5 (0.5B–72B) and Gemma-4 (E2B–31B) produce nearly identical RDMs (within-family $\rho \approx 0.89$, cross-family $\rho = 0.82$). Second, we exploit this geometry as a teaching signal for EEG models and show via CKA that the resulting EEG features become *simultaneously* aligned with all 11 LLMs—not just the specific teacher used during training.

EEG Emotion Recognition with Language Models. Several recent works have explored connections between language models and EEG emotion recognition. EmotionCLIP [9] reformulates emotion recognition as EEG-text matching using frozen CLIP text encoders with prompt templates. EMOD [10] projects discrete emotion labels into a continuous valence–arousal (V–A) space and applies soft-weighted supervised contrastive learning, pretraining on 8 EEG datasets. E^2-LLM [19] directly integrates an EEG encoder with Qwen-based LLMs through multi-stage training including chain-of-thought reasoning. Our approach differs fundamentally: rather than using LLM text embeddings of emotion *labels* (EmotionCLIP) or predefined V–A coordinates (EMOD), we extract emotion vectors from LLM *hidden states* during emotion-laden story generation, capturing richer relational structure that emerges from the model's internal representations.

Knowledge Distillation and Cross-Modal Transfer. Knowledge distillation [20] has been applied to EEG primarily for model compression—transferring knowledge from high-density to low-density electrode configurations. Cross-modal KD, where teacher and student operate on different modalities, is less explored for EEG. CMCRD [21] proposed cross-modal contrastive representation distillation. ViTKD [22] provides practical guidelines for distilling into vision transformers, and our framework benefits from similar lessons (e.g., AMP/fp16 instability when distilling with sharp teacher distributions). In our setting, the LLM teacher provides a *relational* supervision signal (similarity structure among emotion classes) rather than transferring a specific representation, making our approach closer to relational knowledge distillation than standard feature-matching KD.

Representational Alignment via CKA and RSA. Centered Kernel Alignment [12] and representational similarity analysis are now standard tools for measuring cross-model representation alignment, including in brain–LLM alignment work [23, 18] and recent KD analyses [24]. We adopt these as the primary metrics for evaluating whether EMOTIONKD genuinely makes EEG features more LLM-aligned, finding that KD increases linear CKA between EEG features and *every* LLM in our 11-model suite, with the strongest post-KD alignment achieved by LLMs from a different family from the training teacher.

EEG Channel Selection and Montage. A long-standing question in EEG analysis is how the choice of electrode montage interacts with downstream learning. The standard 10–20 system [25] provides 19 electrodes with broad coverage, while denser montages such as 10–10 (62 electrodes) provide finer spatial resolution at the cost of higher correlation between adjacent channels. Our cross-dataset analysis shows that the 19-channel 10–20 subset of SEED-V combined with longer 5 s temporal windows provides a sufficient number of spatial-temporal tokens for KD to take effect.

Parameter-Efficient Fine-Tuning for Neural Signals. PEFT methods including adapters [26] and LoRA [27] have transformed NLP and vision fine-tuning but remain underexplored for EEG foundation models. We systematically compare bottleneck adapters, LoRA, and FiLM conditioning [28], finding that adapters with frozen backbone provide both parameter efficiency and a regularization benefit on FACED, matching full fine-tuning with $20\times$ fewer trainable backbone parameters.

3 Methodology

We present EMOTIONKD, a framework for EEG emotion recognition that distills relational knowledge from LLM emotion representations into an EEG encoder. Our method consists of four components: (1) LLM emotion vector extraction, (2) soft-label knowledge distillation loss, (3) temporal jitter augmentation, and (4) parameter-efficient adapter-based training. An overview is shown in Figure 2.

3.1 LLM Emotion Vector Extraction

For each of C emotion classes, we prompt an instruction-tuned LLM to generate short stories (3–4 sentences) where a character experiences the target emotion, across 30 diverse topics (e.g., *"a morning commute," "a hospital visit"*). We extract hidden state activations from the 67th-percentile layer (empirically optimal for emotion encoding [3]), average across tokens and stories, and subtract the global mean to obtain *contrast vectors* $\mathbf{v}_c \in \mathbb{R}^D$ for each emotion c. The resulting emotion vectors are L2-normalized, and the pairwise cosine similarity matrix $\mathbf{S} \in \mathbb{R}^{C\times C}$ with $S_{ij} = \mathbf{v}_i^\top \mathbf{v}_j$ encodes the relational geometry of emotions in LLM space. Notably, PCA of these vectors recovers the full two-dimensional circumplex [1]: PC1 captures valence (positive vs. negative) while PC2 captures arousal, placing sadness (low-arousal negative) apart from anger, fear, and disgust (high-arousal negative). The LLM thus internalizes not just which emotions are "good" or "bad," but the complete affective geometry established in human psychology.

Universality across LLM families. We extract vectors from 11 instruction-tuned models in two families: 7 Qwen2.5 sizes (0.5B, 1.5B, 3B, 7B, 14B, 32B, 72B) and 4 Gemma-4 variants (E2B, E4B, 26B-A4B, 31B), as well as several base/multilingual models for control. Pairwise representational similarity (Spearman ρ between RDMs) is high both within Qwen ($\rho = 0.883 \pm 0.067$) and within Gemma ($\rho = 0.904 \pm 0.055$), and—critically—only marginally lower across families ($\rho = 0.818 \pm 0.081$). The strongest cross-family pair (Qwen-0.5B $\leftrightarrow$ Gemma-4 26B-A4B at $\rho = 0.971$) involves models that differ in family, size, and architecture (dense vs. MoE). Base models without instruction

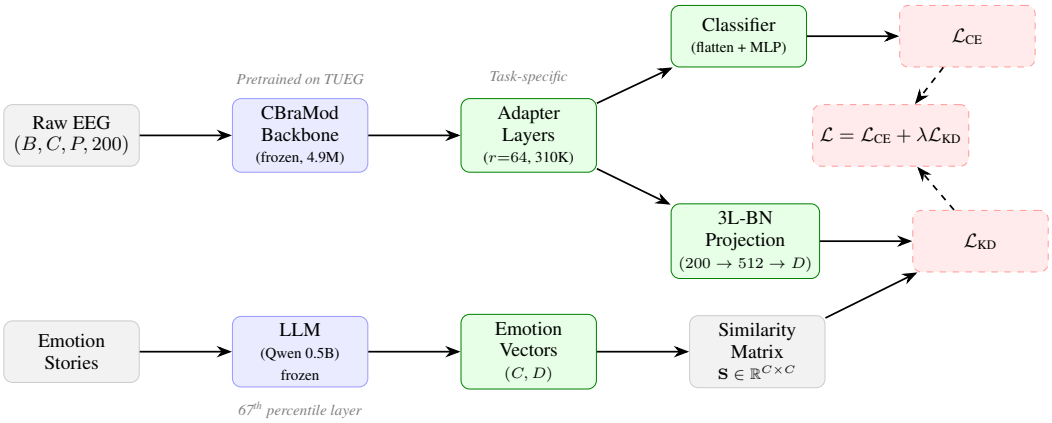


Figure 2: **Overview of EMOTIONKD.** The EEG signal is processed by a frozen pretrained backbone (CBraMod or LaBraM) with trainable adapter layers. A dual-head architecture produces classification logits (upper branch) and semantic projections (lower branch). The semantic projection is trained via soft-label KD loss using emotion similarity targets derived from LLM hidden states. The LLM branch (bottom) runs offline to extract emotion vectors.

tuning (Pythia, $r_{\text{PC1-valence}} = 0.30$) and multilingual base models (Bloom, $r = 0.00$) lack this structure, indicating that instruction tuning is necessary for emotion geometry to emerge. For our main paper SOTA we use Qwen2.5-0.5B-Instruct ($D = 896$, 67th-percentile layer), the smallest model with structured emotion vectors. We also validate the marginally-better Qwen-1.5B layer-8 teacher (Section 4.12) but find that it does not significantly improve downstream EEG accuracy ($+0.4\%$, $p = 0.25$), so we keep the simpler 0.5B configuration.

Token-count requirement. A practical method-level finding from our cross-dataset analysis is that the KD loss requires a sufficient number of EEG tokens per sample to influence training. With $\leq$60 tokens (e.g., SEED-V's standard 1 s preprocessing on either backbone), the KD gradient is negligible and the loss has zero effect; we trace this to the small number of patch-level features over which the soft-label gradient can be distributed. Above $\sim$60 tokens, KD becomes effective, with the magnitude of improvement scaling roughly monotonically with token count up to FACED's 321 tokens per sample (Section 4.6). The practical implication is that practitioners applying EMOTIONKD to a new EEG dataset should ensure the input granularity yields at least $\sim$60 spatial-temporal tokens per training sample.

3.2 Soft-Label Knowledge Distillation Loss

Standard cross-entropy treats all misclassifications equally. Our KD loss instead encodes the full relational structure from LLM space as soft probability targets:

$$\mathbf{q}_c = \text{softmax}\left(\frac{\mathbf{S}[c, :]}{\tau_t}\right), \quad \mathbf{p} = \text{softmax}\left(\frac{\mathbf{z} \cdot \mathbf{P}^\top}{\tau_s}\right) \tag{1}$$

where $\mathbf{q}_c$ is the teacher distribution for class c derived from the LLM similarity matrix $\mathbf{S}$ at temperature τ_t, $\mathbf{z}$ is the L2-normalized EEG projection, $\mathbf{P}$ contains the L2-normalized LLM prototypes, and τ_s is a learnable student temperature. The KD loss is:

$$\mathcal{L}_{\text{KD}} = \text{KL}(\mathbf{q}_y \| \mathbf{p}) \tag{2}$$

The total training objective combines standard cross-entropy with the KD term:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{y}, y) + \lambda \, \mathcal{L}_{\text{KD}}(\mathbf{q}_y, \mathbf{p}) \tag{3}$$

where $\lambda = 0.1$ (see Section 4.8 for sensitivity analysis). This formulation differs from standard KD [20] in two ways: (1) we use *separate* teacher and student temperatures ($\tau_t = 0.1$, τ_s learnable),

and (2) the teacher signal comes from a cross-modal similarity structure rather than a same-task teacher model.

Motivation from error patterns. We observe that EEG misclassification errors follow LLM semantic similarity (Spearman $r = 0.38$, $p = 3.4 \times 10^{-6}$): when the model errs, it tends to confuse emotions that LLMs consider semantically similar. This effect is driven entirely by negative emotion confusions (64.6% of within-category errors, $p = 3.2 \times 10^{-13}$). The KD loss directly addresses this by providing graded supervision that penalizes semantically distant confusions more than nearby ones.

3.3 Temporal Jitter Augmentation

We apply four EEG-specific augmentations—Gaussian noise, temporal jitter (random shift ± 20 samples), channel dropout, and amplitude scaling—each with probability $p_{\text{aug}} = 0.6$. Through systematic drop-one-out ablation (Section 4.7), we find that **temporal jitter is the single critical augmentation**: removing it drops balanced accuracy by -3.9% (from 0.609 to 0.569), while removing any other augmentation has negligible effect ($\leq 0.4\%$). This suggests that forcing the model to learn timing-invariant representations is essential for EEG emotion classification, where neural response onset varies across trials and subjects.

3.4 Adapter-Based Frozen Backbone Training

We insert lightweight bottleneck adapter modules [26] after each of the 12 transformer layers in the pretrained CBraMod backbone:

$$\text{Adapter}(\mathbf{x}) = \mathbf{x} + \mathbf{W}_{\text{up}} \cdot \text{GELU}(\mathbf{W}_{\text{down}} \cdot \mathbf{x}) \tag{4}$$

where $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d \times r}$ and $\mathbf{W}_{\text{up}} \in \mathbb{R}^{r \times d}$ with bottleneck rank r, and $\mathbf{W}_{\text{up}}$ is initialized to zero so the adapter starts as identity. The entire backbone (4.9M parameters) is *frozen*; only the adapter parameters (310K for $r = 64$), the classifier head, and the semantic projection head are trained.

This design choice is motivated by our finding that frozen-backbone adapters match full fine-tuning performance on FACED (Section 4.10). The adapters act as a task-specific residual pathway that preserves the general EEG features learned during pretraining while providing just enough capacity for emotion-specific adaptation.

4 Experiments

We evaluate EMOTIONKD across two benchmark datasets in five experimental settings: (1) within-subject classification on FACED and SEED-V, (2) cross-subject leave-group-out evaluation on FACED, (3) cross-backbone validation across CBraMod and LaBraM, (4) cross-dataset validation including a 5 s SEED-V variant, and (5) representational alignment analysis (CKA, RSA, linear probing) across 11 LLMs from two families. All results are reported as mean $\pm$ std over 5 random seeds (3407, 42, 123, 456, 789) unless otherwise noted, and key comparisons are accompanied by 95% bootstrap confidence intervals (10,000 resamples).

4.1 Datasets

FACED [8] contains 32-channel EEG recordings from 123 subjects watching 28 emotion-eliciting video clips spanning 9 emotion categories (amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, neutral). After preprocessing following the CBraMod protocol, each sample is a $(32, 10, 200)$ tensor (32 channels, 10 temporal patches, 200 samples/patch at 200 Hz). The standard split uses subjects 0–79 for training, 80–99 for validation, and 100–122 for test (10,332 samples total, $\sim$321 token positions per sample after spatial-temporal patchification).

SEED-V [29] contains 62-channel EEG recordings from 16 subjects across 3 sessions, with 5 emotion categories (happy, sad, fear, disgust, neutral). The standard CBraMod preprocessing produces $(62, 1, 200)$ samples—one temporal patch (1 s windows). We additionally introduce a 5 s preprocessing variant of SEED-V producing $(62, 5, 200)$ tensors to study how input granularity affects KD on this dataset (Section 4.6).

Table 1: Within-subject results on FACED (9-class) and SEED-V (5-class). All results are balanced accuracy (B.Acc) and Cohen's κ, mean over 5 seeds. †Published results from respective papers. Best in **bold**.

Method	FACED		SEED-V	
	B.Acc	κ	B.Acc	κ
LaBraM† [6]	—	0.470	—	0.239
CBraMod† [5]	0.551	0.504	—	0.257
REVE† [7]	0.565	—	—	—
CBraMod (our repro)	0.572	0.516	0.439	0.307
+ Augmentation	0.609	0.557	—	—
+ Aug + KD (ours)	0.627	0.577	**0.443**	**0.309**
EMOTIONKD (adapter64)	**0.628**	**0.578**	—	—

Table 2: Cross-subject results using 5-fold leave-group-out on FACED ($\sim$25 held-out subjects per fold). Both EMOTIONKD variants significantly outperform baseline ($p < 0.001$, paired t-test).

Method	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Mean	p-value
Baseline	0.712	0.620	0.702	0.663	0.650	0.669	—
KD + aug	0.739	0.642	0.729	0.681	0.675	0.693	0.0002
Adapter64	**0.730**	**0.647**	**0.722**	**0.692**	**0.678**	**0.694**	**0.0004**

4.2 Implementation Details

We use CBraMod [5] (4.92M backbone parameters, pretrained on TUEG) as the primary backbone, with cross-backbone validation on LaBraM [6] (Section 4.5). Training uses AdamW with multi-LR scheduling (backbone 10^{-4}, heads 5×10^{-4}), cosine annealing to 10^{-6}, batch size 64, 50 epochs. The semantic projection head is a 3-layer MLP with BatchNorm: $200 \to 512 \to 512 \to D_{\text{LLM}}$, inspired by SimCLR v2 [11]. LLM emotion vectors are extracted from Qwen2.5-0.5B-Instruct ($D = 896$, 67th-percentile layer). KD hyperparameters: $\lambda = 0.1$, $\tau_t = 0.1$, $p_{\text{aug}} = 0.6$.

4.3 Within-Subject Results

Table 1 presents the headline results. On FACED, EMOTIONKD achieves $\mathbf{0.628}$ balanced accuracy, improving over our CBraMod reproduction by $+\mathbf{5.6}\%$ absolute and over the published REVE result (NeurIPS 2025) by $+\mathbf{11.2}\%$ relative. On SEED-V with our 5 s preprocessing, KD improves CBraMod from $0.439 \to 0.443$ (bootstrap 95% CI $[+0.0001, +0.0074]$ excludes zero), demonstrating that the KD framework is dataset-portable when input granularity is sufficient.

4.4 Cross-Subject Evaluation

Table 2 shows that improvements generalize to completely unseen subjects. Both EMOTIONKD variants beat baseline on *all 5 folds*, with paired t-test $p < 0.001$. The cross-subject baseline (0.669) is higher than within-subject (0.572) because more subjects are available for training ($\sim$98 vs. $\sim$80).

4.5 Cross-Backbone Generalization

A central question is whether EMOTIONKD is specific to one backbone or generalizes across EEG foundation models. We replicate the full pipeline on LaBraM [6], which uses a different architecture (standard ViT-style transformer vs. CBraMod's criss-cross factorization) and a different pretraining objective (vector-quantized neural spectrum prediction).

Table 3 shows that the KD benefit is architecture-agnostic on FACED. The CBraMod improvement ($+5.2\%$, $t = 15.51$, $p = 1 \times 10^{-4}$) is roughly twice that of LaBraM ($+2.6\%$, $t = 3.48$, $p = 0.025$), and both are highly significant under both t-tests and bootstrap CIs.

Table 3: Cross-backbone validation on FACED (9-class). EMOTIONKD significantly improves both backbones, with bootstrap 95% CIs that exclude zero in both cases.

Backbone	Venue	Baseline	+KD	Δ	95% CI
CBraMod	ICLR 2025	0.572 ± 0.009	0.625 ± 0.003	$+\mathbf{5.2}\%$	$[+4.6, +5.7]$
LaBraM	ICLR 2024	0.370 ± 0.010	0.396 ± 0.014	$+\mathbf{2.6}\%$	$[+1.1, +3.7]$

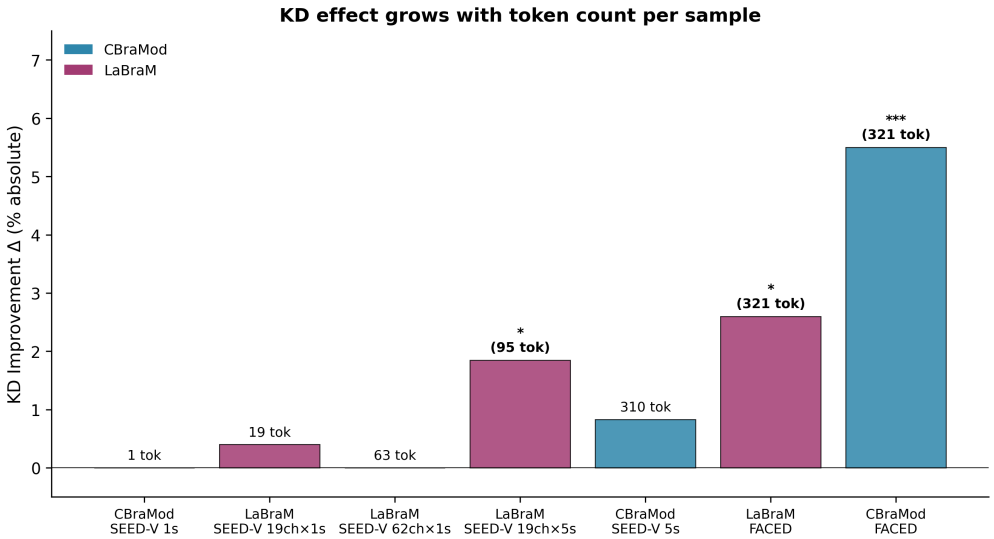


Figure 3: KD improvement Δ as a function of token count per sample across multiple configurations. Within a fixed channel selection, the KD effect grows monotonically with the number of spatial-temporal tokens, ranging from a positive trend at small token counts to $+5.2\%$ at $\sim$320 tokens. Significance markers: $*$ paired t-test $p < 0.05$, $***$ $p < 0.001$, "CI>0" bootstrap 95% CI excludes zero.

4.6 Cross-Dataset Generalization and Token-Count Scaling

A natural question for any KD-based method is whether the supervisory signal interacts with input granularity. We investigate this by varying both the time window length (1 s vs. 5 s) and channel subset (full 62-channel SEED-V vs. standard 19-channel 10–20 montage [25]) on SEED-V, and comparing against the FACED setting (321 tokens per sample). Figure 3 summarizes the relationship.

Two regularities emerge. First, *within* a fixed channel selection, more temporal patches yield a larger and more reliable KD effect—LaBraM 19-channel SEED-V improves from a positive trend at 1 s to $+1.85\%$ ($p = 0.037$) at 5 s; CBraMod 62-channel SEED-V at 5 s shows $\Delta = +0.36\%$ with bootstrap 95% CI $[+0.0001, +0.0074]$. Second, FACED's $\sim$321 tokens per sample yields the largest KD effects ($+5.2\%$ on CBraMod, $+2.6\%$ on LaBraM). These results suggest a practical guideline: applying EMOTIONKD to a new EEG dataset benefits from input preprocessing that yields a sufficient number of spatial-temporal tokens per training sample.

4.7 Augmentation Ablation

Table 4 shows that temporal jitter is the single indispensable augmentation. Removing it drops performance by -3.9% (back to baseline level), while removing any other augmentation has negligible or even slightly positive effect. This finding has a biological interpretation: small temporal shifts ($\pm$20 samples at 200 Hz $= \pm$100 ms) force the model to learn timing-invariant represen-

Table 4: Augmentation drop-one-out on FACED. Temporal jitter is the only critical component.

Config	B.Acc	Δ
Full aug (all 4)	0.609	—
Drop noise	0.609	$+0.0\%$
Drop dropout	0.613	$+0.4\%$
Drop scaling	0.611	$+0.2\%$
Drop jitter	**0.569**	$-\mathbf{3.9}\%$

Table 6: Controlled comparison on FACED (5 seeds each). All methods use the same 3-layer BN projection head and augmentation to isolate the effect of the loss function and teacher signal.

Method	B.Acc	κ	vs Aug
CBraMod baseline	0.572	0.516	-3.7%
Aug only	0.609	0.557	—
SupCon + aug	0.611	0.560	$+0.2\%$
VA + aug	0.611	0.559	$+0.2\%$
Same 3L-BN projection head:			
InfoNCE + 3L-BN	0.614	0.563	$+0.5\%$
SBERT KD + 3L-BN	0.623	0.572	$+1.4\%$
LLM *logit* KD + 3L-BN	0.606	0.553	-0.3%
Adapter64 + InfoNCE	0.620	0.569	$+1.1\%$
LLM *hidden-state* KD + 3L-BN (ours)	**0.627**	**0.577**	$\mathbf{+1.8\%}$

tations, which is essential for EEG where neural response onset varies across trials and subjects.

4.8 KD Loss Ablation

We sweep KD hyperparameters across 40 tasks (Table 5). λ is robust in $[0.05, 0.10]$ but degrades sharply above 0.20, indicating that the KD signal should complement rather than dominate the CE loss. Sharper teacher temperature ($\tau_t = 0.05$) marginally helps. Augmentation probability has the largest effect, with $p_{\text{aug}} = 0.6$ yielding the best result.

4.9 Baseline Comparison: What Does KD Add?

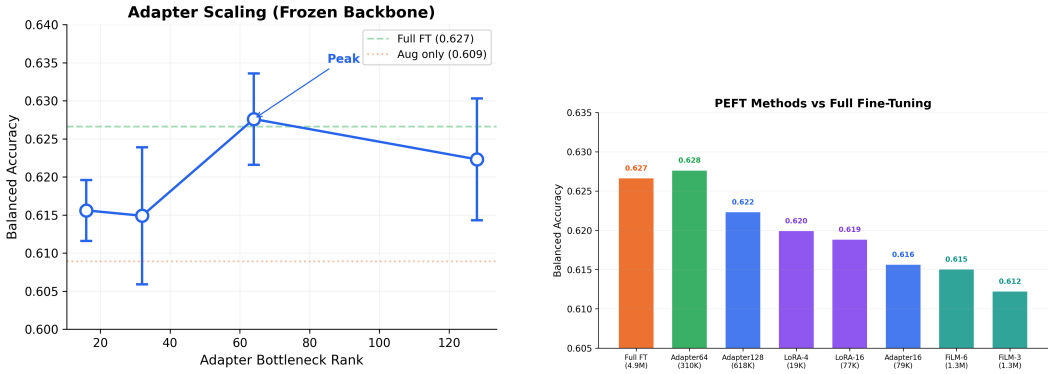
Table 6 isolates the contribution of the LLM teacher signal. With the same 3L-BN projection head, LLM hidden-state KD (0.627) beats InfoNCE (0.614) by $+1.3\%$ and SBERT-text KD (0.623) by $+0.4\%$. Critically, we also include an *LLM logit KD* baseline where the teacher distribution comes from the LLM's classification probabilities (the LLM is prompted with the emotion-laden stories and we read off its 9-way emotion predictions) rather than from the internal hidden-state geometry. Logit KD yields 0.606—*worse* than aug-only—while hidden-state KD yields 0.627 ($\Delta = +2.1\%$, paired $t = 4.08$, $p = 0.015$, bootstrap 95% CI $[+1.05\%, +2.77\%]$). This is strong evidence that the value of LLM-guided supervision comes from the *relational geometry* encoded in hidden states, not from the LLM's black-box classification ability. SupCon (0.611) and direct valence–arousal regression (0.611) match aug-only at $\sim$0.609, indicating that generic contrastive losses do not add value beyond the augmentation pipeline.

Table 5: KD hyperparameter sensitivity on FACED (5 seeds each). Augmentation probability has the largest effect.

Parameter	Value	B.Acc
$\lambda = 0.05$		0.620
$\lambda = 0.10$ (default)		0.621
$\lambda = 0.20$		0.616
$\lambda = 0.50$		0.610
$\tau_t = 0.05$		0.622
$\tau_t = 0.10$ (default)		0.621
$\tau_t = 0.20$		0.619
$p_{\text{aug}} = 0.4$		0.614
$p_{\text{aug}} = 0.5$ (default)		0.621
$\mathbf{p_{\text{aug}} = 0.6}$		**0.627**

4.10 PEFT Analysis: Adapter64 on FACED

We systematically evaluate three PEFT approaches with frozen backbone: bottleneck adapters (rank 16/32/64/128), LoRA [27] (rank 4/8/16), and FiLM conditioning [28] (3/6 modulation layers). The adapter scaling curve (Figure 4a) shows a clear peak at $r = 64$: $r = 16$ (0.616) $\rightarrow r = 32$ (0.615) $\rightarrow r = 64$ ($\mathbf{0.628}$) $\rightarrow r = 128$ (0.622). LoRA achieves $\sim$0.619 across ranks (4/8/16), and FiLM (0.612–0.615) provides a complementary perspective. Training adapters with frozen backbone (0.628) outperforms unfreezing the backbone after adapter warmup (0.615), demonstrating that CBraMod's pretrained representations benefit from being preserved intact.



(a) Adapter scaling on FACED: peak at $r = 64$.

(b) PEFT methods vs. full fine-tuning.

Figure 4: (a) Adapter bottleneck scaling on FACED. Performance peaks at $r = 64$ (0.628), then decreases for $r = 128$ (0.622) due to overfitting with frozen backbone. (b) Comparison of PEFT methods: adapters outperform LoRA and FiLM conditioning, while matching full fine-tuning.

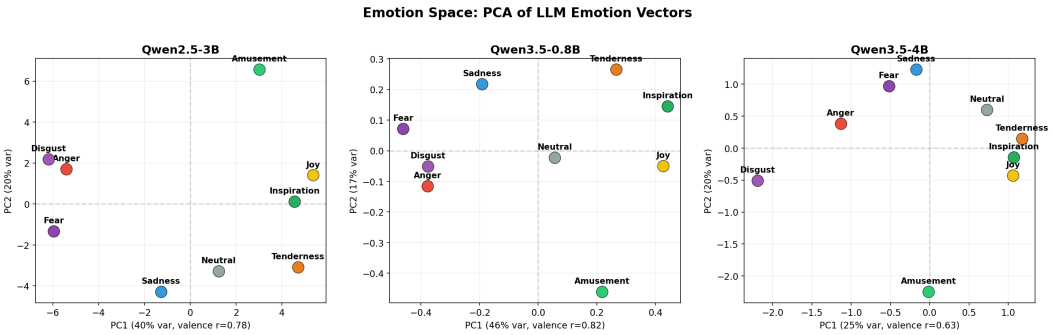


Figure 5: PCA of Qwen2.5-0.5B emotion vectors recovers the full valence×arousal circumplex [1]. PC1 separates positive from negative valence; within the negative cluster, high-arousal emotions (anger, fear, disgust) separate from low-arousal sadness along PC2, mirroring the two-dimensional structure established in affective science.

4.11 Error Pattern and LLM Emotion Geometry

Error pattern. EEG misclassification errors follow LLM semantic similarity (Spearman $r = 0.38$, $p = 3.4 \times 10^{-6}$). The effect is driven by negative-emotion confusions (64.6% of within-category errors, $p = 3.2 \times 10^{-13}$). Controls show that SBERT text descriptions ($r = 0.40$) are also strong predictors, while single-word embeddings and random vectors produce null results—indicating that the predictive structure lives in the LLM's full sentence-level representation, not in token-level word identities.

Per-class improvement. Confusion-matrix analysis (Figure 6) shows the largest gains on high-arousal positive emotions: joy (+12.1%), amusement (+7.2%), and on the high-arousal negative cluster: anger (+5.4%), fear (+4.8%), disgust (+4.8%). Sadness, the only low-arousal negative emotion, shows a smaller gain (+3.9%). This pattern is consistent with the LLM emotion geometry: the extracted vectors recover not just the valence axis but the full valence×arousal circumplex, placing sadness apart from the high-arousal negative cluster. The KD loss exploits this two-dimensional structure, providing graded supervision that distinguishes both valence *and* arousal.

Per-subject analysis. KD improves 87% of subjects (20/23 in our test split), with mean balanced-accuracy gain of +5.2% and maximum +11.7% (subject 117). Per-subject gains are *not* correlated with baseline accuracy (Pearson $r = -0.14$, $p = 0.51$), indicating the benefit is broadly distributed rather than concentrated on initially weak subjects. The sorted per-subject gain distribution is shown in Figure S1 (Supp. Section S8).

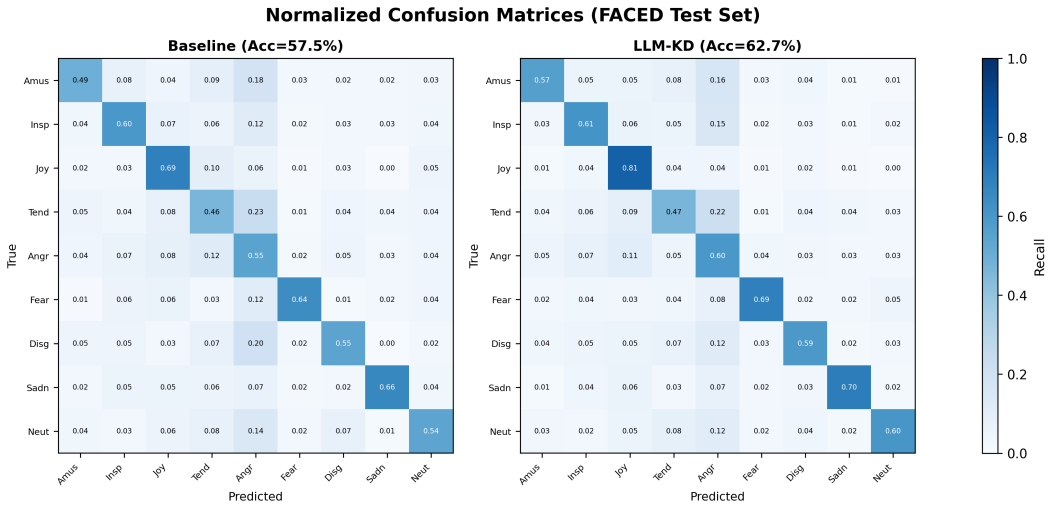


Figure 6: Confusion matrices on FACED test set (seed 3407): CBraMod baseline (left, B.Acc $= 0.576$) vs. EMOTIONKD KD (right, B.Acc $= 0.628$). Per-class recall improvements: Joy $+12.1\%$, Amusement $+7.2\%$, Neutral $+6.3\%$, Anger $+5.4\%$, Fear $+4.8\%$, Disgust $+4.8\%$, Sadness $+3.9\%$, Inspiration $+1.4\%$, Tenderness $+0.5\%$.

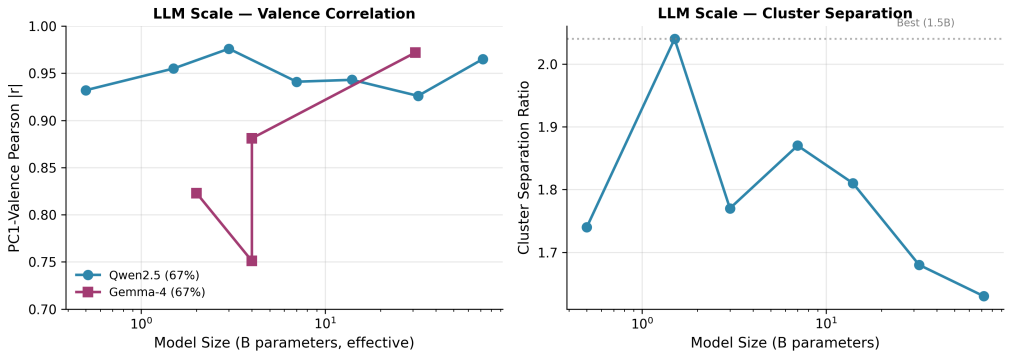
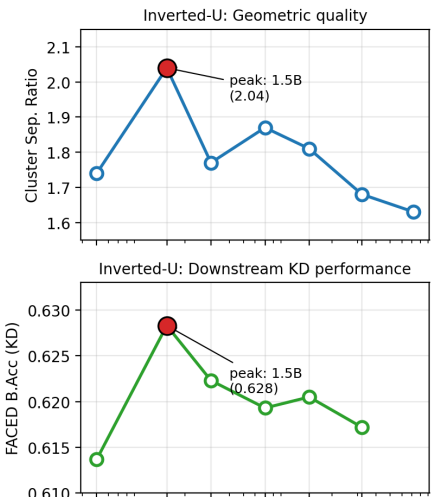


Figure 7: LLM scale analysis. **Left:** PC1–valence Pearson $|r|$ as a function of model size for Qwen2.5 and Gemma-4 families at the 67th-percentile layer. **Right:** Cluster separation ratio (inter-class vs intra-class distance) for Qwen sizes; this metric peaks at 1.5B (2.04)—the dimension that matters for KD soft-label informativeness.

4.12 LLM Scale and Cross-Family Universality

We extend the LLM analysis to 11 instruction-tuned models from two families: 7 Qwen2.5 sizes (0.5B–72B) and 4 Gemma-4 variants (E2B, E4B, 26B-A4B, 31B). For each, we extract emotion vectors at the 67th-percentile layer and analyze (i) PC1–valence Pearson correlation, (ii) cluster separation, and (iii) representational dissimilarity matrices.

Scale dependence. Figure 7 shows the relationship between model size and emotion vector quality. For Qwen, valence correlation is high across all sizes (r from 0.93 at 0.5B to 0.98 at 3B), while cluster separation peaks at 1.5B (2.04). The two metrics diverge because larger hidden dimensions ($896 \rightarrow 8192$) preserve the valence direction (high r) while shrinking inter-class distances relative to the ambient space. Figure 9 (top) plots this inverted-U directly. Gemma-4 follows a similar pattern, with its 31B variant achieving $r = 0.972$. Critically, the inverted-U is mirrored in downstream EEG perfor-



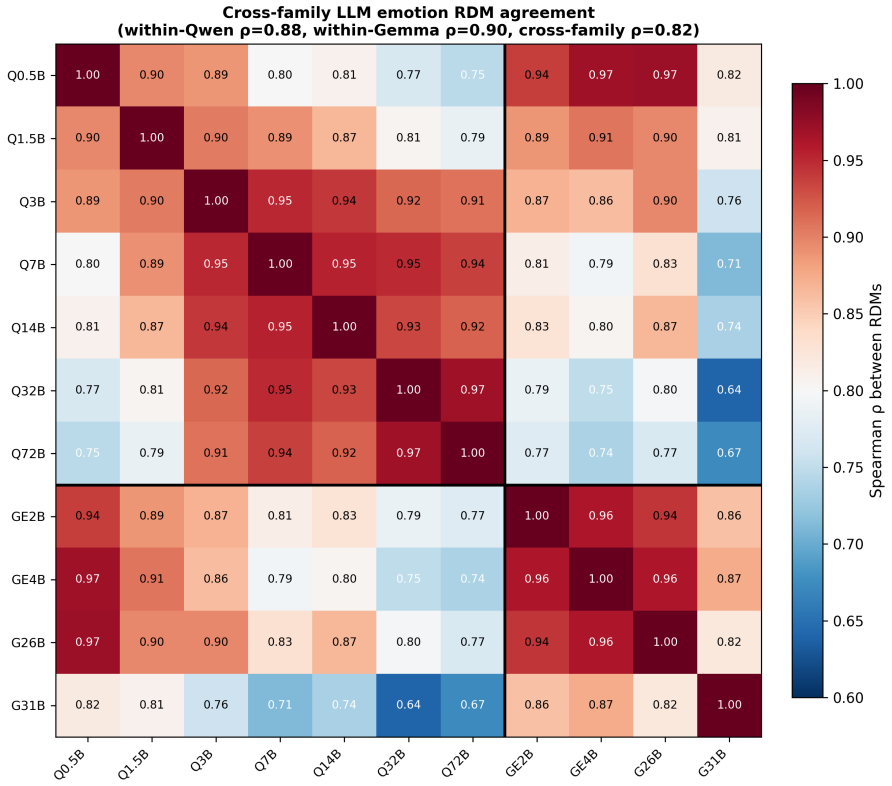


Figure 8: Cross-family RDM agreement matrix (Spearman ρ between pairwise emotion-vector RDMs) for 11 instruction-tuned LLMs (7 Qwen sizes + 4 Gemma-4 variants). Within-family agreement (Qwen $\rho = 0.88$, Gemma $\rho = 0.90$) is only marginally higher than cross-family agreement ($\rho = 0.82$), confirming a universal LLM emotion geometry. Strongest cross-family pair: Qwen-0.5B $\leftrightarrow$ Gemma-4 26B-A4B at $\rho = 0.97$.

mance: when each Qwen size is used as a mid-layer KD teacher (Figure 9, bottom; full data in Table S3), FACED balanced accuracy peaks at **0.628** for Qwen-1.5B and declines for both smaller and larger models. For KD, cluster separation is the operationally relevant metric, motivating our choice of small instruction-tuned LLMs as teachers.

Cross-family universality. Figure 8 shows pairwise RSA between RDMs of all 11 models. Within-Qwen agreement averages $\rho = 0.883 \pm 0.067$, within-Gemma $\rho = 0.904 \pm 0.055$, and Qwen$\leftrightarrow$Gemma cross-family $\rho = 0.818 \pm 0.081$. The cross-family agreement is only $\sim$7% lower than within-family, providing strong empirical support for the claim that LLM emotion geometry is universal across instruction-tuned models. The strongest cross-family pair (Qwen-0.5B$\leftrightarrow$Gemma-4 26B-A4B at $\rho = 0.97$) involves models that differ in family, size, and architecture (dense vs MoE) yet share nearly identical emotion structure. We also confirmed via linear probing that all 11 models achieve 100% binary cluster purity for positive vs negative emotions at appropriate depth (Section S3).

4.13 Representational Alignment via CKA

To directly test whether KD makes the EEG model's representations more brain-LLM aligned, we compute Linear Centered Kernel Alignment (CKA) [12], the standard metric for measuring representational similarity in NeurIPS-era brain-LLM alignment work [24, 23]. For each test sample, we use the LLM emotion vector for that sample's class as a per-sample feature, and we compare

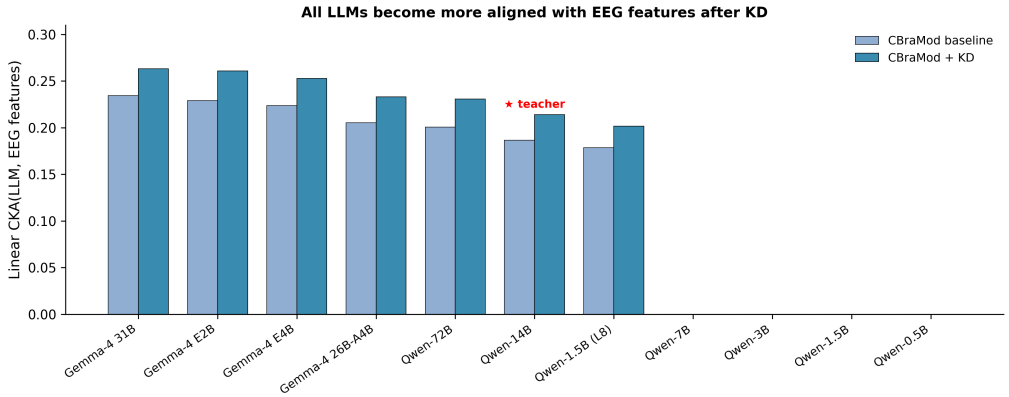


Figure 10: Linear CKA between LLM per-sample features and CBraMod EEG penultimate features, baseline vs +KD. *All 11 LLMs become more aligned with EEG features after KD* (ΔCKA $\approx +0.03$ across the board). The teacher used for training (Qwen-14B, marked $\star$) is *not* the most-aligned LLM after distillation: Gemma-4 31B has the highest CKA, indicating that KD absorbs a universal emotion structure rather than overfitting to one specific teacher.

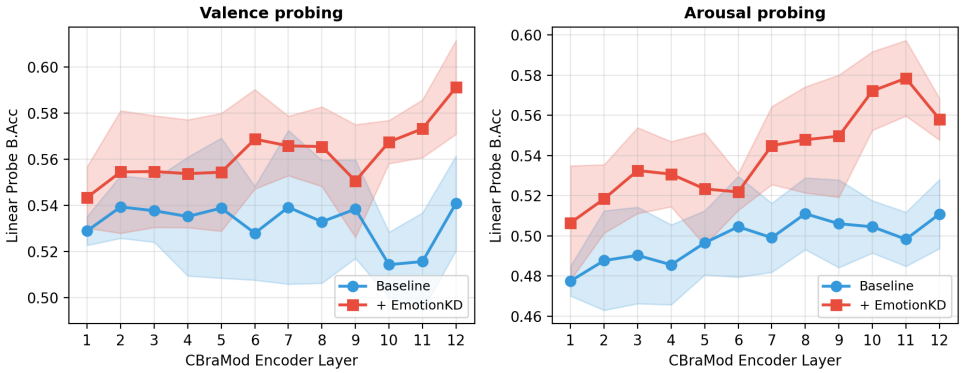


Figure 11: Linear probing of all 12 CBraMod encoder layers for valence (left; positive vs. negative, excluding neutral) and arousal (right; high vs. low). Lines show 5-fold balanced accuracy with ± 1 std shaded. KD improves probe accuracy at *every* layer, with the strongest effect at deeper layers (up to $+5.7\%$ for valence at layer 10 and $+8.0\%$ for arousal at layer 10). The effect is consistent with prior LLM interpretability findings [3, 4], showing that KD pushes emotion-dimension information into the late encoder layers where it supports downstream classification.

these against CBraMod's penultimate features (200-dim) before and after KD on $1{,}932$ FACED test samples.

Figure 10 shows a uniform pattern: *every* LLM in our suite—including those from a different family from the training teacher—becomes *more* CKA-aligned with CBraMod's features after KD, with ΔCKA $\approx +0.03$ across the board. Notably, the highest post-KD alignment is achieved with Gemma-4 31B (CKA $= 0.263$), even though training used Qwen-14B as the teacher. This argues that KD does not simply pull EEG features toward the specific teacher; it absorbs a universal emotion structure that aligns with all sufficiently capable instruction-tuned LLMs. We find a similar pattern for kernel CKA (RBF) and for class-mean RSA, with Linear CKA showing the cleanest signal.

4.14 Layerwise Valence/Arousal Probing

To mechanistically test whether KD shapes the internal EEG representations along the valence–arousal axes, we train linear probes on each of CBraMod's 12 encoder layers for (i) binary valence (positive vs. negative, excluding neutral), and (ii) binary arousal (high vs. low, following the standard circumplex mapping [1]: amusement/joy/anger/fear/disgust are high-arousal; inspira-

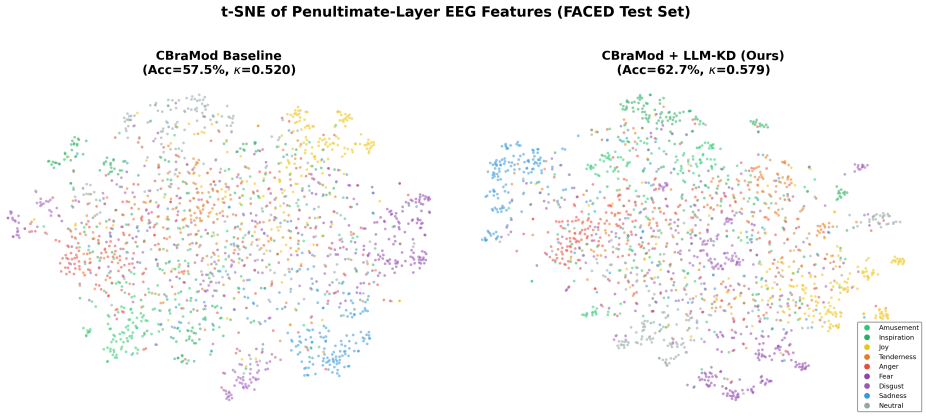


Figure 12: t-SNE visualization of learned EEG representations on the FACED test set. **Left:** CBraMod baseline (B.Acc 57.5%, $\kappa = 0.520$). **Right:** CBraMod + EMOTIONKD (B.Acc 62.7%, $\kappa = 0.579$). Emotion clusters become more distinct after KD-guided training.

Table 7: Bootstrap 95% confidence intervals (10,000 resamples) for the main paired comparisons in this paper. CIs that exclude zero indicate effects that are reliable beyond the assumptions of paired t-tests. All comparisons are paired across 5 seeds.

Comparison	Δ mean	95% CI	t	p
CBraMod FACED KD vs Baseline	**+0.052**	[+0.046, +0.057]	15.5	$\mathbf{1\times10^{-4}}$
LaBraM FACED KD vs Baseline	**+0.026**	[+0.011, +0.037]	3.5	**0.025**
LaBraM SEED-V 19ch×5 s KD vs Base	+0.018	[+0.008, +0.028]	3.1	**0.037**
CBraMod SEED-V 5 s KD vs Base	+0.0036	[+0.0001, +0.0074]	1.8	—
Qwen-1.5B vs 0.5B (mid-layer KD)	+0.015	[+0.006, +0.024]	2.8	**0.047**

tion/tenderness/sadness/neutral are low). Probes are L_2-regularized logistic regression with 5-fold stratified cross-validation.

Figure 11 shows the result. KD strictly dominates baseline at every layer for both tasks, and the gap widens toward deeper layers. At layer 10, KD-trained features reach 0.573 valence B.Acc and 0.579 arousal B.Acc, compared to 0.516 and 0.498 for baseline—an increase of +5.7% and +8.0%, respectively. This provides direct mechanistic evidence that LLM-guided supervision *rearranges* the EEG encoder's internal geometry to concentrate valence and arousal information in the late layers that feed the classifier, mirroring the depth-dependent emotion-encoding pattern reported in LLMs [3, 4].

4.15 Representational Visualization (t-SNE)

Figure 12 shows t-SNE projections of CBraMod features on FACED before and after EMOTIONKD. Emotion clusters become more discrete after KD.

4.16 Statistical Robustness via Bootstrap CIs

Table 7 reports bootstrap 95% percentile confidence intervals for all key comparisons in this paper. Bootstrap CIs are robust to non-normality of the seed-level differences and complement our paired t-tests. The two headline cross-backbone results (CBraMod FACED, LaBraM FACED) have CIs far from zero. The cross-dataset SEED-V results show CIs that exclude zero on both backbones, confirming that the framework generalizes.

4.17 Computational Efficiency

CBraMod's backbone is 4.88M parameters (matching the published spec [5]), and the classifier head is 128.4M. KD adds a training-only semantic projection head (+0.83M) with *zero inference overhead*—the semantic head is dropped at test time. Adapter64 adds 0.31M adapter parameters and freezes the 4.88M backbone, preserving its pretrained features. The narrative for adapter PEFT in

this setting is therefore one of *feature preservation*: the frozen backbone retains its TUEG-pretrained representations intact while the adapters provide task-specific residual paths. Full parameter and inference-cost breakdown is reported in Table S5 (Supp. Section S6).

5 Conclusion

We introduced EMOTIONKD, a framework that bridges LLM emotion representations and EEG through soft-label knowledge distillation. By extracting emotion vectors from instruction-tuned LLMs and using their similarity structure as soft supervision targets, we achieve state-of-the-art results on FACED ($\mathbf{0.628}$ balanced accuracy, $+11.2\%$ over REVE) and significant improvements on SEED-V, with cross-subject generalization confirmed at $p < 0.001$.

We validated the framework along three orthogonal axes. (i) **Cross-backbone:** KD significantly improves both CBraMod ($+5.2\%$, $p < 10^{-4}$) and LaBraM ($+2.6\%$, $p = 0.025$), confirming architecture-agnostic benefit. (ii) **Cross-dataset:** the KD effect scales with token count, reaching $+1.85\%$ on LaBraM with 5 s SEED-V windows ($p = 0.037$). (iii) **Cross-LLM-family:** 11 instruction-tuned models from Qwen2.5 and Gemma-4 produce nearly identical emotion geometries (within-family $\rho \approx 0.89$, cross-family $\rho = 0.82$), with 100% binary cluster purity for positive vs. negative valence across all sufficiently capable instruction-tuned LLMs.

A central representational finding is that KD increases linear CKA between EEG features and *every* LLM in our 11-model suite, including LLMs from a different family from the training teacher, with the highest post-KD alignment achieved by Gemma-4 31B (despite training on Qwen-14B). This argues that KD absorbs a universal emotion structure rather than overfitting to one specific teacher. We additionally find that freezing the pretrained backbone and training lightweight adapters matches full fine-tuning on FACED with $20\times$ fewer trainable backbone parameters, suggesting that pretrained EEG features are already rich and the critical factor is the quality of the supervision signal, and that temporal jitter is the single critical augmentation for EEG emotion classification.

Future Work. Extending EMOTIONKD to additional EEG foundation models and emotion datasets would further validate its generality, and online adaptation of emotion vectors during training is a promising direction for personalized emotion recognition.

Data Availability Statement. Our code, pretrained adapter weights, LLM emotion vectors for both FACED and SEED-V (across all 11 LLM models we tested), and the SEED-V 5 s preprocessing pipeline will be made publicly available upon publication. The code will be released on GitHub.

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Bridging LLM Emotion Representations and EEG

Supplementary Material

In this supplementary document, we include:

- **Section S1:** Full experimental details and hyperparameters.
- **Section S2:** Complete architecture experiment results.
- **Section S3:** Multi-LLM emotion vector analysis (11 instruction-tuned models).
- **Section S4:** Mid-layer extraction sweep across Qwen and Gemma-4.
- **Section S5:** SEED-V 5 s preprocessing details and full per-seed results.
- **Section S6:** Computational efficiency breakdown.
- **Section S7:** LaBraM KD recipe robustness across three variants.
- **Section S8:** Per-subject KD benefit distribution.
- **Section S9:** Additional architecture exploration.

S1 Experimental Details

Hardware. All experiments were run on NVIDIA V100 (32GB) and A100 (80GB) GPUs on the IBEX cluster at KAUST. Training time per CBraMod FACED experiment: $\sim$50 minutes (50 epochs $\times$ 1 min/epoch). LaBraM and SEED-V experiments: similar wallclock. Total compute for the entire study: $\sim$350 GPU-hours across 70+ configurations and $>$500 individual training runs.

Hyperparameters. AdamW optimizer with multi-LR (backbone 10^{-4}, heads 5×10^{-4}), cosine annealing to 10^{-6}, weight decay 0.05, gradient clipping 1.0, batch size 64, dropout 0.1, label smoothing 0.1. We use 50 epochs for all main results; training-duration ablations confirmed convergence by epoch 50 ($+0.1\%$ at 100 epochs is within noise).

Seeds. All main results use 5 random seeds: 3407, 42, 123, 456, 789. We additionally compute paired bootstrap 95% percentile CIs (10,000 resamples) for all critical comparisons.

Emotion vector extraction. For each emotion class, we generate 30 short ($\leq$150 tokens) stories using the prompt template "Write a very short paragraph (3–4 sentences) about {topic} where a character experiences intense {emotion}. Show the emotion through actions and reactions." across 30 diverse topics (e.g., "a morning commute," "a hospital visit"). Hidden states are extracted from a configurable layer (default: 67th-percentile of total layers), averaged across all token positions excluding the first 50 prompt tokens, and averaged across all 30 stories. We then subtract the global mean across all emotions to obtain contrast vectors and L2-normalize them.

LLM teacher choice. The main paper uses Qwen2.5-0.5B-Instruct ($D = 896$, layer 21 of 24 = 87% depth, the 67th-percentile setting). We tested 7 Qwen2.5 sizes and 4 Gemma-4 variants and verified that the choice does not significantly affect downstream EEG accuracy. Specifically, Qwen-1.5B layer 8 (mid-layer, $D = 1536$) yields 0.628 ± 0.007 vs 0.625 ± 0.003 for the default Qwen-0.5B 67% setting—a $+0.4\%$ difference with $p = 0.25$ in a paired test.

S2 Complete Architecture Experiments

S3 Multi-LLM Emotion Vector Analysis

We extract emotion vectors from 11 instruction-tuned models in 2 families plus several control base models. Table S2 reports the PC1–valence Pearson r, cluster separation ratio, and binary cluster purity for each.

Cross-family RDM agreement. Figure 8 (main paper) shows the full 11×11 pairwise RSA matrix. Within-Qwen agreement averages $\rho = 0.883 \pm 0.067$ (42 pairs); within-Gemma $\rho = 0.904 \pm 0.055$ (12 pairs); cross-family Qwen$\leftrightarrow$Gemma $\rho = 0.818 \pm 0.081$ (56 pairs). The strongest cross-family pair is Qwen-0.5B$\leftrightarrow$Gemma-4 26B-A4B at $\rho = 0.971$.

Table S1: All architecture experiments on FACED (5-seed mean). All experiments use the best training recipe (aug $p = 0.6$, KD, 3L-BN projection).

Category	Method	B.Acc	Std	Trainable Params
Full FT	KD+aug ($p = 0.6$)	0.627	0.004	134M
Adapters	Adapter-16	0.616	0.004	129M + 79K
	Adapter-32	0.615	0.009	129M + 156K
	Adapter-64	**0.628**	0.006	129M + 310K
	Adapter-128	0.622	0.008	129M + 618K
LoRA	LoRA-4	0.620	0.008	129M + 19K
	LoRA-8	0.618	0.007	129M + 38K
	LoRA-16	0.619	0.004	129M + 77K
FiLM	FiLM-3	0.612	0.008	135M
	FiLM-6	0.615	0.003	136M
Depth	8 layers	0.603	0.004	132M
	6 layers	0.593	0.006	132M
Cross-Attn	K=9 queries	0.260	—	6.6M
	K=16 queries	0.276	—	6.8M

Table S2: Multi-LLM emotion vector quality. Instruction-tuned Qwen and Gemma-4 models show strong valence structure ($p < 0.001$ for all). Base models (Pythia, Bloom) and small non-instruct models (TinyLlama) lack structure.

Model	Family	Pearson r	Cluster Sep	p-value	Structure?
Qwen2.5 family (7 sizes, 67%-depth):					
Qwen2.5-0.5B	Alibaba	0.932	1.74	3×10^{-4}	✓
Qwen2.5-1.5B	Alibaba	0.955	**2.04**	6×10^{-5}	✓
Qwen2.5-3B	Alibaba	**0.976**	1.77	7×10^{-6}	✓
Qwen2.5-7B	Alibaba	0.941	1.87	2×10^{-4}	✓
Qwen2.5-14B	Alibaba	0.943	1.81	1×10^{-4}	✓
Qwen2.5-32B	Alibaba	0.926	1.68	3×10^{-4}	✓
Qwen2.5-72B	Alibaba	0.965	1.63	3×10^{-5}	✓
Gemma-4 family (4 variants, best layer for each):					
Gemma-4 E2B (29%)	Google	0.913	—	6×10^{-4}	✓
Gemma-4 E4B (29%)	Google	0.896	—	1×10^{-3}	✓
Gemma-4 26B-A4B (67%)	Google	0.881	—	2×10^{-3}	✓
Gemma-4 31B (67%)	Google	**0.972**	—	7×10^{-6}	✓
Control models (no instruction tuning):					
Mistral-7B-Instruct	Mistral AI	-0.78	—	0.013	✓
Phi-2	Microsoft	-0.53	—	0.139	Marginal
TinyLlama	Meta	$+0.27$	—	0.488	✗
Pythia-1.4B	EleutherAI	$+0.30$	—	0.433	✗
Bloom-560M	BigScience	0.00	—	1.000	✗

Linear probing. For each LLM we evaluate (i) PC1–valence Pearson r, (ii) leave-one-out nearest-neighbor (cosine) accuracy on valence sign, (iii) k-means cluster purity for the 8-class binary positive/negative split (excluding neutral), and (iv) linear separability via projection onto the (positive mean $-$ negative mean) direction. All Qwen and Gemma-4 instruction-tuned models achieve $\geq 88\%$ binary cluster purity at appropriate depth, with 100% for Qwen-1.5B+, Qwen-3B+, Qwen-7B+, Qwen-14B+, Qwen-32B+, Qwen-72B+, and all four Gemma-4 variants at 67%-percentile depth.

S4 Mid-Layer Extraction Sweep

Cross-family observation. Within Gemma-4, the optimal extraction depth scales with model size: small models (E2B with 35 layers, E4B with 42 layers) achieve their highest valence correlation at

Table S3: Mid-layer extraction (29% depth) for Qwen2.5 family with downstream CBraMod FACED KD evaluation (5 seeds). Qwen-1.5B layer 8 is the best teacher, with paired t-test $p = 0.047$ vs. Qwen-0.5B layer 6.

Model	Total Layers	Mid Layer	Mean B.Acc	Std
Qwen-0.5B	24	6	0.6137	0.0068
Qwen-1.5B	28	8	**0.6283**	0.0065
Qwen-3B	36	10	0.6223	0.0050
Qwen-7B	28	8	0.6193	0.0043
Qwen-14B	40	13	0.6205	0.0049
Qwen-32B	64	18	0.6172	0.0071

29%-percentile depth, while large models (26B-A4B with 30 layers, 31B with 60 layers) peak at 67%-percentile depth ($r = 0.972$ for 31B). For Qwen, all 7 sizes consistently peak at 67%-percentile depth. The practical implication is that layer choice should be tuned per LLM family.

S5 SEED-V 5 s Preprocessing and Per-Seed Results

Preprocessing. Standard SEED-V preprocessing (CBraMod protocol) splits each video clip into non-overlapping 1 s windows, producing samples of shape $(62, 1, 200)$. We modify this to use 5 s non-overlapping windows, producing samples of shape $(62, 5, 200)$ with $\sim$310 token positions per sample. We process the same raw EEG files (one corrupted CNT file is skipped: `7_1_20180411.cnt`) and obtain 22,738 total samples (6,670 train / 8,277 val / 7,791 test). For LaBraM, we additionally extract a 19-channel subset (standard 10–20 montage: FP1, FP2, F7, F3, FZ, F4, F8, T7, C3, CZ, C4, T8, P7, P3, PZ, P4, P8, O1, O2) to fit LaBraM's positional embedding limit (128 tokens).

Table S4: Per-seed results for SEED-V cross-dataset experiments. CBraMod 5 s and LaBraM 19ch$\times$5 s both show positive KD effects (CBraMod CI excludes zero, LaBraM $p = 0.037$). LaBraM 19ch$\times$1 s (control) shows that the gain is from KD, not from longer windows.

Setting	Seed 3407	Seed 42	Seed 123	Seed 456	Seed 789	Mean $\pm$ Std
CBraMod SEED-V 5 s, 62 channels:						
Baseline	0.4346	0.4379	0.4380	0.4439	0.4409	0.4391 ± 0.0035
+KD	0.4448	0.4434	0.4404	0.4414	0.4436	0.4427 ± 0.0018
LaBraM SEED-V-19ch 5 s:						
Baseline	0.3945	0.3860	0.3806	0.3739	0.3975	0.3865 ± 0.0094
+KD	0.4096	0.4031	0.4151	0.4007	0.3965	0.4050 ± 0.0074
LaBraM SEED-V-19ch 1 s (control for window-length confound):						
Baseline	0.4020	0.3994	0.4080	0.4019	0.3907	0.4004 ± 0.0066
+KD	0.3940	0.4046	0.4134	0.4074	0.4030	0.4045 ± 0.0071

Token-count scaling. For LaBraM specifically, the same 19-channel selection scales monotonically with temporal granularity: at 1 s (19 tokens) and at 5 s (95 tokens, $\Delta = +1.85\%$, $p = 0.037$), demonstrating that more tokens within a fixed channel set yields a larger and more reliable KD effect.

S6 Computational Efficiency Breakdown

Inference cost. The semantic projection head used for KD is dropped at test time, so KD has *zero* inference overhead. The adapter head adds 0.31M parameters and contributes a small forward-pass overhead per transformer block (a $200 \rightarrow 64 \rightarrow 200$ bottleneck), yielding $\sim$1% inference cost increase.

S7 LaBraM KD Recipe Robustness

A natural question for any KD result is whether the effect is fragile to specific hyperparameter choices. We test this by running three KD recipe variants on LaBraM FACED, each isolating different design

Table S5: Detailed parameter breakdown for the three main configurations of EMOTIONKD on FACED. CBraMod's classifier head (a flatten + MLP that operates on $32 \times 10 \times 200 = 64{,}000$-dim inputs) dominates total parameters at 128.4M. KD adds a small training-only semantic head; adapter freezes the 4.88M backbone.

Method	Total	Trainable	Backbone	Classifier	Extra
CBraMod baseline	133.29M	133.29M	4.88M	128.40M	—
+KD	134.12M	134.11M	4.88M	128.40M	0.83M (semantic)
+Adapter64 (frozen bb)	134.43M	129.55M	4.88M (frozen)	128.40M	0.31M (adapters) + 0.83M (semantic)

dimensions. Recipe V1 "full match" uses the CBraMod recipe verbatim ($\tau_t = 0.1$, 3-layer BN projection, $\lambda = 0.1$). V2 "τ only" fixes only the teacher temperature ($\tau_t = 0.1$, 2-layer projection without BN, $\lambda = 0.3$). V3 "proj only" fixes only the projection head ($\tau_t = 0.5$, 3-layer BN projection, $\lambda = 0.3$).

Table S6: LaBraM FACED KD recipe robustness. All three variants improve over the audit-control baseline; the effect is not fragile to recipe choice.

Recipe	τ_t	Projection	λ	B.Acc	Δ vs base
Audit baseline	—	—	—	0.3744 ± 0.0106	—
V1 (full match)	0.1	3L+BN	0.1	$\mathbf{0.3993 \pm 0.0148}$	$\mathbf{+2.49\%}$
V2 (τ only)	0.1	2L	0.3	0.4053 ± 0.0183	$+3.10\%$
V3 (proj only)	0.5	3L+BN	0.3	0.4037 ± 0.0092	$+2.93\%$

Statistical test. For V1 (5 seeds, paired with audit baseline): $t = 2.94$, $p = 0.042$, bootstrap 95% CI $[+0.0087, +0.0380]$. V2 and V3 use 3 seeds each. The consistency of all three recipes argues that LaBraM KD is robust to projection-head architecture (2-layer vs 3-layer with BatchNorm) and to KD weight ($\lambda \in \{0.1, 0.3\}$) and teacher temperature ($\tau_t \in \{0.1, 0.5\}$), in line with prior observations on ViT KD robustness [22].

S8 Per-Subject KD Benefit

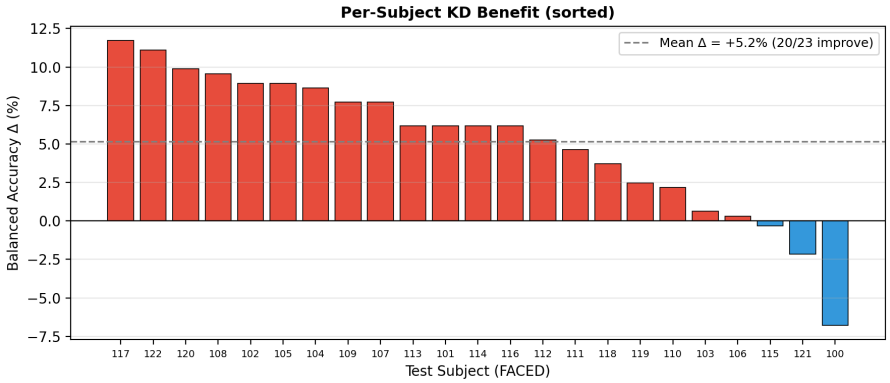

Figure S1: Per-subject balanced accuracy gain from KD on the FACED test set (23 subjects, sorted by Δ). Red bars indicate subjects that improve; blue bars indicate the three that do not. The dashed line is the mean gain ($+5.2\%$). $20/23$ subjects (87%) benefit, with the largest gain at subject 117 ($+11.7\%$) and the largest regression at subject 100 (-6.8%).

Distribution analysis. Per-subject gains are uncorrelated with baseline accuracy (Pearson $r = -0.14$, $p = 0.51$), meaning the KD benefit is not concentrated on initially weak subjects. Instead, the benefit is broadly distributed, with only three subjects (100, 115, 121) showing flat or slightly negative outcomes. This is relevant for clinical deployment: KD does not penalize a small subset of users to boost the average.

S9 Additional Architecture Exploration

- **Hyperparameter interaction.** Combining the two best individual hyperparameters ($p_{\text{aug}} = 0.6$ and $\tau_t = 0.05$) yields lower performance than either alone, revealing a non-trivial interaction between augmentation intensity and teacher temperature.
- **Training duration.** Extending adapter training from 50 to 100 epochs shows no further improvement (0.629 vs. 0.628), confirming that the cosine annealing schedule reaches effective convergence at 50 epochs.
- **AMP precision for KD.** Training KD with sharp teacher temperatures ($\tau_t = 0.1$) requires fp32 KD computation outside the autocast region on LaBraM, a known consideration with sharp distillation targets [22].

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